

Key: Building a Score From Scratch

84-355 Democracy's Data | Session 12, part 2 — the Supreme Court's votes | Instructor
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All code is given to students. Grade the interpretation.

Structure. The brief (`scdb-brief.Rmd`) is a chapter, “Forty-Four Numbers and a Guess”. It is organised by argument under prose headings and carries four numbered figures — Figure 1, the agreement matrix in both orders, with the toggling D3 panel; Figure 2, the scaling with the guess and without it; Figure 3, majority rates against position; Figure 4, the by-term series. It no longer has numbered steps. The lab (`scdb.Rmd`) runs the same sequence in Parts 1–6 with the code exposed and ends with four options. **Students should read the brief first. The Step numbers below are this key's own walking order for the session,** the eleven rows of the timing table, and the lab's Parts map onto them (Part 1 = Step 1, Part 2 = Steps 2 and 6, Part 3 = Steps 3–7, Part 4 = Step 8, Part 5 = Step 10, Part 6 = the options). Where the brief is meant it is named, and cited by figure number or heading.

This is the second half of Session 12. The morning lab (`dw-nominate`) hands students a published ideal-point score and asks them to read it. This one builds one in front of them, from 44 numbers, and then breaks it. The pairing is the point: the same method, once as a consumer and once as a manufacturer.

Every number below was computed from the files in `data/` at knit time. **The brief computes its numbers the same way; the lab does not.** `scdb.Rmd` prints live output but states the figures in prose by hand — 96.2%, -0.975, 0.224, and the rest. They are all correct against the current files, checked. If `data/build-data.R` is ever re-run against a newer SCDB release, **the lab's prose is the thing that goes stale**, and this key and the brief will not warn you, because they will simply report the new numbers.

The session in one sentence

Feed nothing but “how often did these two vote the same way” into a one-dimensional scaling and the conventional left-right ordering of the Roberts Court falls out — but one of the 45 cells contains no data, and what you put there decides the fine ordering, so the blocs are in the data and the ranking is in the assumptions.

The findings

```

c(justices      = NJ,
  pairs         = NPAIR,
  pairs_with_data = NPAIR - nrow(HOLE),
  fill_value    = round(FILL, 3),
  cor_with_fill  = round(R, 3),
  cor_refusing   = round(R2, 3),
  bloc_span      = round(SPAN, 3),
  chasm          = round(CHASM, 3),
  positions_moved = MOVED,
  one_vote_cases = nrow(cc),
  bloc_only_wins  = BLOC_WINS,
  terms         = nrow(bt))

```

```

#>      justices      pairs pairs_with_data      fill_value      cor_with_fill
#>      10.000      45.000      44.000      0.732      -0.975
#>      cor_refusing      bloc_span      chasm positions_moved      one_vote_cases
#>      -0.990      0.145      0.224      4.000      30.000
#>      bloc_only_wins      terms
#>      10.000      78.000

```

```

# all nine adjacent gaps; the fill only ever decides the two smallest
round(GAPS, 4)

```

```

#>      SAAIto      NMGorsuch      ACBarrett      JGRoberts      BMKavanaugh      KBJackson
#>      0.0004      0.0855      0.0114      0.0466      0.0009      0.2239
#>      EKagan      SGBreyer      SSotomayor
#>      0.0544      0.0251      0.0155

```

```

# how often the recovered order survives resampling the votes

```

```

c(exact_order_pct = SAME,
  alito_before_thomas_pct = FTA,
  kavanaugh_before_roberts_pct = FRK)

```

```

#>      exact_order_pct      alito_before_thomas_pct
#>      5.65      49.45
#>      kavanaugh_before_roberts_pct
#>      48.25

```

- The ordering is **Thomas, Alito, Gorsuch, Barrett, Roberts, Kavanaugh, then a gap, then Jackson, Kagan, Breyer, Sotomayor** — recovered from 44 agreement rates with no ideology supplied.
- It **correlates -0.975** with each justice's share of conservative votes, a variable the scaling never saw. Refusing to guess, on 8 justices: **-0.990**.
- It is **two clusters, not a spectrum**. The six on one side span 0.145; the single step across the middle is **0.224**.
- **One cell of 45 has no data**. Jackson and Breyer never sat together. Filling it with the mean moves 4 positions.
- **And those positions were never determined anyway**. The two pairs that swap are separated by 0.0004 and 0.0009 on a scale spanning 0.464. Resample the votes and the full order comes back **5.7% of the time**; Alito and Thomas trade places on essentially a coin flip.
- **Kavanaugh is in the majority 95.5% of the time** and Roberts 94.3%. In the 30 one-vote decisions both fall to **73.3%**.

- **Only 10 of the 30 one-vote decisions** were won by a majority drawn entirely from the six-justice bloc the scaling found. Two thirds of the closest cases cut across the line the lab just built.
- The long series runs **78 terms, 1946 to 2023**, every value a coder’s judgement about direction.

The three things to get said

1. The blocs are in the data; the fine ordering is in the assumptions. This is the pivot. Steps 6 and 7 are the session. A student who leaves knowing only that one empty cell reorders the justices has got the course’s transferable lesson: **every published score has a fill**, a cell with nothing in it and a decision about what to put there, and the question to ask of any such number is not whether it made assumptions but which ones and what happens if you change them. Make somebody state the asymmetry out loud — the coarse structure survives refusing to guess (-0.990 on 8 justices, blocs unchanged) and the fine structure does not.

And go one step past the brief. The two pairs that swap are separated by 0.0004 and 0.0009. They were never distinguished in the first place. The fill is not overturning a finding; it is breaking a tie that the data never broke. That is worse than it sounds and better than it sounds at the same time, and saying which is the discussion.

2. “Moderate” is a description of the winning side, not of anyone’s opinions. Step 8 lands hard: the two justices nearest the middle of the recovered line win 95.5% and 94.3% of cases, and the ones at the ends lose a quarter to a third. It is a mechanical consequence of position, not a personality trait. **Then take it back with Step 9:** restricted to the 30 one-vote decisions those same two fall to 73.3%, and only 10 of the 30 were won by the conservative bloc alone. The 95.5% figure is mostly a fact about unanimous and lopsided cases, which are most cases.

3. Every number in the 78-term series is a human coding decision, and every row of the underlying file is a person’s act. The direction codes are a codebook applied case by case since 1946. That is not a reason to discard the series; it is a reason to say what it is. And note the level: this is **individual-level data** — Thomas’s vote is Thomas’s vote, not a rate computed over a group — which is why the agreement matrix is even possible. Aggregate data cannot support this analysis at all.

The line for the board: the score is a measurement of nine people and a measurement of one decision we made, and the arithmetic cannot tell them apart.

Walking the steps

Step	What it does	Min	Watch for
1 What one record is	Ten justices, cases, direction, majority	5	The two short tenures are planted here. Point at them; do not explain yet.
2 Who agrees with whom	45 pairs, 96.2% to 53.0%	6	“No ideology has been supplied to it” is the sentence that has to land.

Step	What it does	Min	Watch for
3 The score falls out	<code>cmdscale</code> on 1 minus agreement	7	Students think a model was fitted. Nothing was fitted; distances were arranged. The sign is arbitrary (see below). Do not let anyone read meaning into it. Ask for the prediction <i>before</i> showing the chasm. Undefined, not zero, not missing. No future data fixes it. The pivot. Take the prediction in the block quote seriously — most rooms guess “nothing changes” or “everything changes”, and both are wrong. Someone will call it power. It is geometry. Denominators differ by justice — see the caution below. This is Step 6’s problem wearing a different hat. The series stops at 2023. Say so out loud. Push for “the blocs are in the data” in a student’s own words.
4 Check against something unseen	Correlation -0.975	4	
5 Not a spectrum	Span 0.145 vs chasm 0.224	5	
6 One cell is not a number	Jackson and Breyer, 0 shared cases	8	
7 Refuse to guess	4 positions move; blocs do not	12	
8 The number that surprises	95.5% in the majority	7	
9 Close cases	30 one-vote decisions	10	
10 Seventy-eight terms	1946–2023, coded direction	8	
11 What you can now say	And cannot	6	

Total 78 minutes; Steps 7 and 9 absorb overruns.

If you are short on time, protect Steps 6, 7 and 9. The undefined cell, what happens when you refuse to fill it, and the collapse of the majority rates in close cases are the argument. Steps 1–5 can be walked in ten minutes as a narrated demonstration; Step 10 can be a single sentence over the plot.

What the brief now shows before any of this

Figure 1 of the brief, under “Who agrees with whom”, draws all 45 agreement rates as a 10 x 10 heatmap, with the one undefined pair left blank and outlined in red, and a button labelled *re-sorted by the recovered order* (two panels in the PDF) that re-sorts the same matrix. **Use it as the cold open.** Students see the block structure and the hole with no method in between, which makes Steps 3 to 6 an explanation of something they have already noticed rather than a sequence of results they have to take on trust. Figure 3, under “Does the middle still win when it is close?”, adds the majority rates plotted against position: an arch for all cases, and a flatter, partly inverted shape for the one-vote cases.

Two cautions about the numbers themselves

The axes have no meaning and no orientation

Classical multidimensional scaling recovers **distances**, not coordinates. A scaling solution can be reflected, rotated, or both without changing its fit by any amount at all: if you multiply every position by -1 the pairwise distances are identical, so the fit is identical. **“Left” and “right” on that plot are conventions the analyst imposes after the fact, not findings.**

Three practical consequences:

- The correlation with conservative voting is -0.975 **and it is negative only because the solution came out with conservatives on the negative side.** A reflected solution would give 0.975. The magnitude is the finding; the sign is not.
- **The brief now states this under “The score falls out”,** immediately after the recovered ordering, and both figures that use the scale label the axis as arbitrary in direction. If a student quotes the brief at you, it agrees with this page.
- The lab’s plot is labelled **position** (left = conservative here). The word “here” is doing real work. **If the plot ever renders mirrored on a student’s machine, nothing is wrong** — eigen-decomposition sign conventions can differ. Say this before somebody reports a bug.
- In two dimensions (Option A) rotation is free as well, so “the second dimension is X” is a claim about a coordinate system you chose, not about the Court.

This is not a defect peculiar to our version. It applies to Martin-Quinn, to NOMINATE, and to every ideal-point estimate students will ever read about. The orientation is always supplied by the analyst.

The close-case denominators are not all 30

`close_cases.csv` records the term, so each justice’s real denominator is the one-vote decisions in the terms they actually sat for.

```
d <- jj[order(-jj$close_pct), c("justice", "close_maj", "close_n", "close_pct")]
d$justice <- nice(d$justice); d$close_pct <- pc(d$close_pct)
print(d, row.names = FALSE)
```

```
#>   justice close_maj close_n close_pct
#> Kavanaugh      22      30      73.3
#>   Roberts      22      30      73.3
#>   Breyer       12      18      66.7
#>   Jackson       7      12      58.3
#> Sotomayor      17      30      56.7
#>   Kagan        16      30      53.3
#>   Barrett     14      30      46.7
#>   Alito        14      30      46.7
#>   Thomas       13      30      43.3
#>   Gorsuch      13      30      43.3
```

Breyer sat for 18 of the one-vote decisions and Jackson for 12. Dividing either by 30 — which is what a student will do — gives 40.0% and 23.3%, both badly wrong. Barrett, by contrast, was confirmed before the first one-vote decision in the window was handed down, so her 14 of 30 is comparable even though she has fewer coded cases than the rest.

This is the single most likely arithmetic error in Option C, and catching it is a 4.

Option A — Two dimensions

Verified output

```
round(sol$points[order(sol$points[, 1]), ], 4)
```

```
#>           [,1]      [,2]
#> CThomas      -0.2177 -0.0764
#> SAAlito       -0.2173 -0.0047
#> NMGorsuch     -0.1317 -0.0520
#> ACBarrett    -0.1203  0.0563
#> JGRoberts    -0.0737  0.0411
#> BMKavanaugh -0.0728  0.0459
#> KBJackson     0.1511  0.1162
#> EKagan        0.2055 -0.0104
#> SGBreyer      0.2306 -0.1124
#> SSotomayor    0.2461 -0.0035
```

```
round(sol$eig[1:5], 4) # scree: how much each axis carries
```

```
#> [1] 0.3160 0.0418 0.0368 0.0193 0.0061
```

```
round(c(eig2_over_eig1 = sol$eig[2] / sol$eig[1],
        eig2_over_eig3 = sol$eig[2] / sol$eig[3],
        gof_1d = cmdscale(as.dist(1 - M), k = 1, eig = TRUE)$GOF[1],
        gof_2d = sol$GOF[1]), 3)
```

```
#> eig2_over_eig1 eig2_over_eig3      gof_1d      gof_2d
#>           0.132           1.135      0.702      0.795
```

```
round(c(median_cor_dim1_under_resampling = D1COR,
        median_cor_dim2_under_resampling = D2COR), 3)
```

```
#> median_cor_dim1_under_resampling median_cor_dim2_under_resampling
#>                               0.995                               0.685
```

Dimension 1 carries an eigenvalue of 0.316; dimension 2 carries 0.042 and dimension 3 carries 0.037. **The second dimension is barely larger than the third**, which is the cleanest available signature of noise: a real second dimension stands apart from the ones below it, and this one does not. Resampling the votes, the recovered second dimension correlates a median 0.685 with the one in the data — dimension 1 comes back at 0.995 by comparison.

Who it separates matters too. On dimension 2 the extremes are Jackson and Breyer — which are exactly the two justices joined by the invented cell from Step 6. **The second dimension in the ten-justice solution is substantially manufactured by the fill.** Run it on the eight overlapping justices and a different second dimension appears, with Gorsuch at one end and Roberts and Kavanaugh at the other:

```
round(sol2$points[order(sol2$points[, 1]), ], 4)
```

```
#>           [,1]      [,2]
#> SAAAlito      -0.1794  0.0160
#> CThomas       -0.1677 -0.0565
#> NMGorsuch     -0.0768 -0.1251
#> ACBarrett     -0.0596  0.0184
#> BMKavanaugh  -0.0312  0.0839
#> JGRoberts     -0.0285  0.0979
#> EKagan        0.2531 -0.0089
#> SSotomayor    0.2900 -0.0257
```

```
round(sol2$eig[1:4], 4)
```

```
#> [1] 0.2197 0.0368 0.0167 0.0071
```

3/4: Runs $k = 2$, plots both, reports that the second dimension does not change the left–right ordering and separates one or two justices from the rest.

4/4: Names who moves and asks whether it is real. Any of these is the move: notices the second dimension's extremes are Breyer and Jackson and connects that to the fill; or compares eigenvalue 2 to eigenvalue 3 and concludes they are indistinguishable; or proposes a substantive reading (Gorsuch on criminal procedure and Indian law is the standard one) **and says what evidence would distinguish it from noise.**

4/4, best: Says that the question as posed cannot be settled from the agreement matrix at all, because the axes of a scaling solution are only defined up to rotation — dimension 2 is whatever is left over after dimension 1, in a coordinate system nobody chose on substantive grounds. To interpret it you would need to leave the matrix and go look at which cases those justices split on. **Knowing the data cannot answer it is the right answer**, and it is the same move rewarded in Option B of the precinct-geography lab.

Option B — Refuse to guess

Verified output

```
data.frame(rank = seq_along(ord2),
           filled = nice(keep),
           refusing = nice(names(ord2)),
           changed = ifelse(keep == names(ord2), "", "swapped"))
```

```
#>   rank  filled refusing changed
#> 1    1   Thomas    Alito swapped
#> 2    2    Alito   Thomas swapped
#> 3    3  Gorsuch  Gorsuch
#> 4    4  Barrett Barrett
#> 5    5 Roberts Kavanaugh swapped
#> 6    6 Kavanaugh Roberts swapped
#> 7    7    Kagan    Kagan
#> 8    8 Sotomayor Sotomayor
```

```
round(c(cor_with_fill = R, cor_refusing = R2), 3)
```

```
#> cor_with_fill cor_refusing
#>      -0.975      -0.990
```

```
round(ord[keep], 4) # positions with the guess
```

```
#>      CThomas      SAAalto      NMGorsuch      ACBarrett      JGRoberts BMKavanaugh
#>      -0.2177      -0.2173      -0.1317      -0.1203      -0.0737      -0.0728
#>      EKagan      SSotomayor
#>      0.2055      0.2461
```

```
round(ord2, 4) # positions refusing to guess
```

```
#>      SAAalto      CThomas      NMGorsuch      ACBarrett BMKavanaugh      JGRoberts
#>      -0.1794      -0.1677      -0.0768      -0.0596      -0.0312      -0.0285
#>      EKagan      SSotomayor
#>      0.2531      0.2900
```

Two adjacent pairs trade places, 4 positions in all. Nothing else moves, the blocs are identical, and the correlation with conservative voting **improves** from -0.975 to -0.990.

3/4: Re-runs it, reports that two pairs swapped and that the blocs held.

4/4: Draws the general conclusion — coarse structure is robust, fine ordering is not — and says what that implies for reading any published ranking of the justices. Bonus for noticing that the correlation went *up* without the guess, and for resisting the conclusion that the eight-justice version is therefore “better”: it answers a different question, about a Court that never existed.

4/4, best: Notices *which* pairs swapped and checks the gaps. Thomas and Alito are 0.0004 apart and Roberts and Kavanaugh 0.0009, on a scale spanning 0.464 — so the fill did not overturn a result, it broke a tie. A student who says “those two were never distinguished by this data in the first place, with or without the guess” has understood the method better than the brief states it. The resampling above puts a number on it: the full order survives 5.7% of the time and Alito precedes Thomas in 49.5% of resamples.

(That resampling treats each pair’s agreement count as an independent binomial, which it is not — the same votes enter many pairs — so read it as illustrative rather than as a confidence interval. A proper resample would be at the case level and the aggregated files do not support one. Worth saying if a student asks; do not require it.)

Option C — The close cases

Verified output

```
o <- jj[order(-jj$close_pct),
      c("justice", "pct_in_majority", "close_maj", "close_n", "close_pct")]
o$justice <- nice(o$justice); o$close_pct <- pc(o$close_pct)
print(o, row.names = FALSE)
```



```
#> justice pct_in_majority close_maj close_n close_pct
#> Kavanaugh          95.5         22      30      73.3
#> Roberts            94.3         22      30      73.3
#> Breyer             72.1         12      18      66.7
#> Jackson            78.3          7      12      58.3
#> Sotomayor          70.7         17      30      56.7
#> Kagan              74.0         16      30      53.3
#> Barrett           90.9         14      30      46.7
#> Alito              82.0         14      30      46.7
#> Thomas             79.3         13      30      43.3
#> Gorsuch            81.3         13      30      43.3
```

```
# how many one-vote majorities came entirely from the six-justice bloc
table(conservatives_in_majority = nc)
```

```
#> conservatives_in_majority
#>  2  3  4  5
#> 14  4  2 10
```

3/4: Counts correctly, reports that Kavanaugh and Roberts are still top but much lower — 73.3% against 95.5% — and that the ordering below them scrambles.

4/4: Handles the denominator (see the caution above): either restricts to the justices who sat the whole window, or uses each justice’s own terms. A student who divides Breyer’s 12 by 30 and reports 40.0% has made exactly the Step 6 error in a new place, and should be told so rather than marked down hard — it is the most instructive mistake in the lab.

4/4, best: Notices that only 10 of the 30 one-vote decisions were won by a majority drawn entirely from the conservative bloc, and draws the conclusion: **the dimension the scaling recovered explains the Court’s agreement in general but not its closest cases.** Two thirds of the one-vote decisions cut across it. That is a genuine limit on what an ideal-point score buys you, and it is not stated anywhere in the brief — reward it warmly.

Option D — Your own question

No fully verified output; grade the reasoning and the fit between question and data.

3/4: Asks a question the four files can actually answer and answers it without error — most-agreeing pair by term, the shape of the by-term series around a particular appointment, which justices appear together most often in one-vote majorities.

4/4: States what would have to be true for the answer to mean what the student wants it to mean, and checks at least one of those things. The commonest good version: a claim about change over time that runs into the fact that this is a snapshot and the by-term series measures a different quantity (case direction) from the scaling (justice agreement).

4/4, best: Asks a question the data *cannot* answer and explains precisely why — then says what data would. Examples that have worked: “did Roberts move after 2018?” (needs a dynamic model, which is exactly why Martin-Quinn exists); “do these blocs disagree about outcomes or about reasons?” (the file has no opinions in it); “is the docket selected by the same nine people whose agreement we are measuring?” (yes, and it makes the denominator endogenous). **Rejecting a question for a stated reason is worth as much as answering one.**

Common failures: comparing “% conservative” across distant terms without any argument about the code-book; reading a trend off the raw grey line in Step 10 rather than the smoothed one; treating the 78-term series and the four-term scaling as if they measured the same thing.

Grading

Four-point scale from the syllabus.

- **Never penalise code — it is given.** A wrong number from correctly running the given code is a reading error, not a coding error, and should be graded as interpretation.
- **The move that earns a 4** is treating the fill as a substantive decision with consequences rather than a technical detail, and being explicit about which parts of the answer survive changing it.
- **A 2** is “the score is subjective so it doesn’t mean anything.” True in the trivial sense, useless, and the exact conclusion Step 7 is designed to prevent — the whole point is that *some* of it survives and you can say which.
- **Also a 2:** reporting the recovered ranking as though the positions were measured quantities, e.g. “Thomas is the most conservative justice” as a finding rather than as a coin flip with Alito.
- **Also a 2:** dividing every close-case count by 30 and not noticing.
- **Reward anyone who asks how precise the positions are.** The lab gives no error bars and there is no honest way to attach them without going back to the case-level data. Noticing their absence is the most sophisticated available response, and it is what the resampling above exists to answer for you rather than for them.

Anticipated questions

- “*Isn’t this circular? You got the answer everyone already knew.*” — The strongest thing you can say is that the ordering was not put in: the scaling saw 44 numbers and no labels, and the dimension it found is the one the culture already names. The strongest objection is that agreement and ideology may be the same phenomenon measured twice, in which case -0.975 is a tautology with extra steps. **Both are defensible and neither is refutable from this table.** Say that; do not resolve it.
- “*Why not just use Martin-Quinn?*” — Because a published score is the answer, and this session is about how the answer gets made. Also, practically, the Martin-Quinn site is behind a bot check and cannot be fetched by a script; the build note in `data/build-data.R` says so.
- “*Why is conservative negative?*” — It is not. See the orientation caution above. The solution could be reflected at no cost.
- “*Couldn’t you just leave the missing cell out?*” — Not in classical scaling, which needs a complete distance matrix. Option B’s answer is to drop the two justices instead, which is the honest version of leaving it out and costs you two justices.
- “*Why does the data stop at 2023?*” — Because the SCDB release used here is the 2024 release, covering terms through 2023. **Nothing after the 2023 term is in these files**, so no claim in this lab is about the current Court. Anyone reaching for a 2025 or 2026 decision is off the edge of the data.
- *Is Allen v. Milligan, which is sitting in close_cases.csv, still good law?* — Largely not, and it is a gift of an example. *Allen v. Milligan*, 599 U.S. 1 (2023), appears in the file as a 5–4 decision whose majority was **Roberts and Kavanaugh joined with Kagan, Sotomayor and Jackson** — one of the 20 one-vote cases that cut across the recovered line. Then came *Louisiana v. Callais*, 608 U.S. ____ (2026), decided April 29, 2026, in which the Court substantially reworked Voting Rights Act § 2 and held that § 2 did not require Louisiana to draw an additional majority-minority district. Alito wrote for a six-justice majority that included **both Roberts and Kavanaugh**, with Kagan dissenting for herself, Sotomayor and Jackson. **Two of the justices who made the Milligan majority were**

on the other side three years later — a concrete demonstration of the lab’s own limitation that a snapshot cannot show anyone changing. If you use this, get the neighbours right too: *Shelby County v. Holder*, 570 U.S. 529 (2013), struck the § 4(b) coverage formula, not § 5 itself, and *Bartlett v. Strickland*, 556 U.S. 1 (2009), is a plurality, not a majority.

- “*Do the justices know these numbers exist?*” — Yes. Several have commented on them, usually unfavourably. That is a good thirty seconds, not more.

Connections

- **dw-nominate** — the other half of Session 12 and the direct pair. There students consume a published score; here they build one. **Run them in that order**, and ask at the end of this one what DW-NOMINATE’s fill is.
- **levels-of-aggregation** — this lab sits on the **individual** side of the course’s individual-versus-aggregate axis. A justice’s vote is a person’s act, recorded as such, and that is precisely why a pairwise agreement matrix can exist at all; no aggregate table of “the Court voted conservative 60% of the time” could support it. Worth one sentence at the top of the session.
- **cast-vote-records** — the other lab built on individual acts rather than rates, and the closest analogue: an anonymous ballot there, a named vote here. The contrast between the two is a good exam question.
- **rpv** — where the *absence* of individual-level data forces inference from aggregates, and everything gets harder. The cleanest possible foil for this session.
- **precinct-geography** — the same shape of lesson in a different domain: two defensible methods, one invisible choice, a moved answer.
- **surveys-source** and **gss-confidence** — the other places in the course where a headline number turns out to rest on a coding decision made by somebody the reader never meets.